

SECTION 9: ROLES OF GOVERNMENT

9.1 EXTERNALITIES

Multiple Choice Questions

1985/AL/1/02

A chemical company drops waste into a farmer's water supply, causing significant damage to the farmer. The farmer cannot legally ask for compensation from the chemical company. In view of this situation

- A. an output tax should be imposed on the chemical company because of the implied inefficiency.
- B. it is in the farmer's interest to pay the chemical company to decrease waste dumping.
- C. the law should be changed to allow the farmer to demand compensation from the chemical company.
- D. the chemical company should be allocated elsewhere.

1985/AL/1/23

A divergence between private cost and social cost means

- A. the market fails to maximize welfare.
- B. income distribution requires improvement.
- C. resources allocation requires improvement.
- D. the cost to an individual of an action differs from the cost to the society.

1986/AL/1/06

A divergence between private and social costs

- A. is inefficient and should be eliminated by government regulations.
- B. can be resolved by private contracting.
- C. means that income distribution is more unequal.
- D. (out of syllabus)

1986/AL/1/25

The smoke from a metal processing factory in Hong Kong pollutes a large government housing estate nearby, and the toxic nature of the smoke may threaten the lives of residents who are scheduled to move into the estate soon. Economic analysis tells us that

- A. the polluting factory must be relocated.
- B. the residents must not be allowed to move into the housing estate.
- C. life is invaluable and must be saved, therefore either the factory or the housing estate must be relocated.
- D. to maximize social gain there is not enough information to decide on the proper course of action.

1988/AL/1/17

A restaurant opens near a housing estate. The cooking smell annoys the residents nearby. Economic analysis indicates that

- A. the residents of the housing estate should be encouraged to sue the restaurant owner.
- B. resource allocation will be more efficient if the restaurant owner pays the residents to tolerate the smell.
- C. it is more beneficial to society if the government forbids the restaurant to operate near the housing estate.
- D. the mere presence of the annoying smell does not necessarily represent inefficient use of resources.

1990/AL/I/25

Suppose a factory pollutes the environment. If the right to pollute is well-defined and transferable,

- A. the level of pollution is subject to market contracting.
- B. pollution will be eliminated.
- C. the factory will necessarily increase its production.
- D. then, on efficiency grounds, the government should intervene when the right to pollute is not transacted.

1990/AL/I/30

A divergence between private and social costs is **NOT** wasteful if

- A. the majority of people benefit from the divergence.
- B. the cost of eliminating the divergence is greater than the cost of the divergence itself.
- C. people who suffer from the divergence choose not to complain to the government.
- D. the divergence is not expected to last for a long time.

1991/AL/I/01

A divergence between private and social costs

- A. occurs when one person's action inflicts damage on another person.
- B. cannot be resolved without government intervention.
- C. may be eliminated by market transactions if the action that causes the damage is a private property right.
- D. means inefficient allocation of resources.

1992/AL/I/22

Which of the following implies a divergence between private and social costs?

- A. A factory pollutes the neighbouring housing for workers, and this housing is owned by the factory.
- B. Increasing textile exports to the USA leads to higher clothing prices in Hong Kong.
- C. Mr. Smith plays the piano which his neighbour loves to hear, but he plays it so often that the marginal benefit of his music to the neighbour is exactly zero.
- D. None of the above.

1992/AL/I/23

Suppose the construction of a flyover causes the value of nearby properties to fall. To attain economic efficiency,

- A. the flyover builder should compensate the nearby property owners.
- B. the flyover builder should install a noise reduction device.
- C. the nearby building should be relocated.
- D. There is not enough information to answer the question.

1992/AL/I/24

If private ownership of a forest is abolished, and the forest is subject to common exploitation,

- A. the rate of depletion will remain unchanged.
- B. the trees will become a public good.
- C. the private cost of cutting trees will differ from the social cost.
- D. the trees will become a free good because they no longer have a price.

1993/AL/I/08

In its production process, a chemical factory upstream pollutes a river nearby without considering the damages inflicted upon a fish farmer downstream. Under zero cost of transaction,

- A. the chemical factory will pollute more if it has the private property rights to use the river.
- B. there will be no water pollution if the fish farmer has the private property rights to use the river.
- C. there will be no water pollution if the fish farmer purchases the ownership of the chemical factory.
- D. water pollution will be reduced if private property rights over the use of the river are established, regardless of who owns these rights.

1994/AL/I/08

If all property rights are well defined and transaction costs are zero,

- A. there will be no pollution.
- B. the marginal social cost of pollution will equal the marginal social benefit of pollution.
- C. the level of pollution will be minimized.
- D. the existing level of pollution will remain unchanged.

1994/AL/I/15

A divergence between private and social costs results from

- A. market failure.
- B. property rights not being clearly defined.
- C. too little government intervention.
- D. the inability to distinguish private goods from public goods.

1995/AL/I/23

The construction of a new cross harbour tunnel in Western District has caused the value of nearby properties to rise. Therefore, to increase efficiency,

- A. nearby property owners should share the capital gain with the tunnel company.
- B. the tunnel company should ask nearby property owners to share the tunnel construction costs.
- C. the government should encourage property developers to build more new flats in that area.
- D. None of the above.

1996/AL/I/21

Given that people have well-defined rights to enjoy fresh air, which of the following justifies the government prohibiting smoking in public areas?

- A. Smokers inflict harm on non-smokers.
- B. The cost of contracting between smokers and non-smokers is prohibitive.
- C. The rights of non-smokers must be protected at all costs.
- D. It is impossible to find a tax schedule to correct the external effects of smoke pollution.

1997/AL/I/27

A pianist plays the piano. His neighbour enjoys the music but the pianist receives no payment from the neighbour. A divergence between private and social costs/benefits occurs in the following situations **EXCEPT**:

- A. The more music the pianist plays, the more the neighbour enjoys the music.
- B. The more music the pianist plays, the less the neighbour enjoys the music.
- C. The neighbour feels that the last minute of the music has a zero value.
- D. Although the neighbour loves the music, he feels that it would be better if the pianist plays a little less.

1997/AL/I/28

In the United States, as in some other developed countries, one who builds a highrise blocking the view of others may be subject to public hearings, and through these hearings the right to build the highrise may be denied. In Hong Kong, however, so long as the building code is met, every builder has the right to block the view of the residents nearby. Which of the following is true?

- A. A divergence between private and social costs is more likely to occur in Hong Kong.
- B. A divergence between private and social costs is less likely to occur in Hong Kong.
- C. The transaction costs involved in the elimination of the divergence between private and social costs is higher in Hong Kong.
- D. None of the above.

1998/AL/I/13

A factory dumps waste into a farmer's water supply, causing significant damage to the farmer. With prohibitive transaction costs in making contracts,

- A. the situation may be efficient and no government intervention is needed.
- B. the situation is inefficient and the factory should not be allowed to produce.
- C. the farmer should have the right to demand compensation from the factory.
- D. the situation is inefficient and government regulation is the only way to improve efficiency.

1998/AL/I/19

A divergence between private cost and social cost

- A. will not exist if transaction costs are zero.
- B. cannot be resolved without private contracting.
- C. implies that income distribution is unequal.
- D. means that the market is inefficient and requires government regulation.

1999/AL/I/19

The new airport in Hong Kong relieves the people living under the old flight path from noise pollution, but causes noise pollution to those under the new flight path. After the opening of the new airport,

- A. the value of properties under the new flight path will fall if the noise pollution is fully anticipated.
- B. the value of properties under the new flight path will rise if the noise pollution is fully anticipated.
- C. the respective property values will change if changes in the levels of noise pollution are not correctly anticipated.
- D. the respective property values will not change under any circumstances.

1999/AL/I/27

Which of the following implies a divergence between private and social costs?

- A. Male students contribute money towards buying perfumes for female students.
- B. A student accidentally pours coffee on a professor's jacket and says sorry.
- C. A husband complains about his wife all the time but refuses to get a divorce.
- D. A piano player who never receives any payment for his music from his neighbour always plays the piano that the marginal value of his piano play to his neighbour is zero.

2000/AL/I/19

Newspaper and magazine publishers often increase their sales and incomes by reporting personal information about famous movie stars who usually do not receive any financial return from the publishers. However, these movie stars often benefit from these reports which serve as a kind of publicity for them. Which of the following statements is correct regarding this situation?

- A. There must be a divergence between private and social costs.
- B. To attain economic efficiency, the government should force the party who benefits more from these reports to compensate the other party.
- C. The marginal benefits of these reports to the two parties must be the same since neither party pays the other.
- D. The costs of making and enforcing contracts are prohibitively high.

2000/AL/I/20

When a pianist plays the piano up to a point where the marginal value of the unpaid music to his/her neighbour is zero,

- A. private marginal benefit diverges from social marginal benefit.
- B. the total gain to the pianist and his neighbour is maximized.
- C. the property rights to the music has to be defined for economic efficiency.
- D. the time spent on playing the piano is indeterminate.

2003/AL/I/27

The construction of a new airport generates unpleasant noise for nearby residents, and yet the value of nearby properties increases as a result of the new airport. Which of the following is correct?

- A. There may be no divergence between private and social costs.
- B. The existence of both noise damage and increases in property value calls for a tax on the one hand and a subsidy on the other.
- C. There must be a divergence between private and social costs.
- D. There is definitely no divergence between private and social costs.

2005/AL/I/29

The introduction of a new airport near a city as a rule creates noise and pollution that affect the market value of the property nearby. Also as a rule, the market value of the nearby property rises because of a substantial increase in business activities. Which of the following is correct?

- A. There is a divergence between private and social costs, and taxes imposed on airplane take-offs and landings will reduce this divergence.
- B. There is a divergence between private and social costs, and subsidies awarded to airplane take-offs and landings will reduce this divergence.
- C. So long as the market value of nearby property continues to rise with more airplanes using the airport, hence more noise and pollution, the divergence between private and social costs declines.
- D. There is no divergence between private and social costs as long as the market value of nearby property rises.

2006/AL/I/29

Two adjacent houses have trees extending into each other's garden area. Both owners clear the fallen leaves in their own lawns without saying anything to one another. Which of the following statements is correct?

- A. There may be no divergence between private and social costs.
- B. The marginal damage of these trees to each party must be the same since no compensation is given.
- C. There must be a divergence between private and social costs.
- D. The cost of negotiation between the two parties is irrelevant.

2006/AL/I/30

A divergence between private and social costs

- A. may not exist even in the absence of private property rights.
- B. will never exist with private property rights.
- C. will never exist with government intervention.
- D. may still exist even if transaction costs are zero.

2007/AL/I/30

Building a highway creates a nuisance for the environment nearby. Land prices near the highway often increase after the announcement of the route. This implies that

- A. The government has selected the location of the highway accurately.
- B. The benefits derives from the highway outweigh the nuisance created.
- C. Pollution is inevitable when the economy develops.
- D. There is necessarily a divergence between private and social costs.

2008/AL/I/28

In many countries, residents in apartments are not allowed to keep dogs, but residents in garden houses are allowed to do so. Which of the following statements is correct?

- A. The nuisance of dogs barking is transacted in the market.
- B. The divergence between private and social costs increases whenever dogs are allowed to be kept.
- C. The divergence between private and social costs decreases whenever dogs are not allowed to be kept.
- D. None of the above.

2008/AL/I/30

The strict enforcement of a no-smoking law

- A. will eliminate the divergence between private and social costs with respect to the effects generated by smoking in public.
- B. will reduce the divergence between private and social costs of public smoking.
- C. may increase the divergence between private and social costs of public smoking.
- D. is more efficient than the absence of this law because the poll shows that the majority of people support its passing.

2009/AL/I/28

A divergence between private and social costs

- A. may exist even if transaction cost is zero.
- B. may exist even if all resources are allocated by using market prices.
- C. implies the efficiency condition is violated.
- D. implies the absence of private property rights.

2010/AL/I/28

In 1920, A.C. Pigou of Cambridge advanced the argument that, with a superior road and an inferior road, car drivers flock to use the superior road and cause traffic congestion.

- A. There is a divergence between private and social costs in this situation.
- B. A tax imposed by the government on the use of the superior road will reduce the divergence between private and social costs.
- C. If the roads are privately owned, a higher toll will be charged for the use of the superior road.
- D. All of the above

2012/AL/I/28

A.C. Pigou observed that the sparks of passing trains harmed the crops of farmland near a railway. However, the noise of the trains also scared away the birds feeding on the crops and because of this the price of the farmland rose. Which of the following statements is correct?

- A. A tax imposed on the use of the railway would reduce the divergence between private and social costs.
- B. A subsidy granted to the farmers would reduce the divergence between private and social costs.
- C. There is no divergence between private and social costs when the price of nearby farmland reached a maximum.
- D. So long as the market value of nearby farmland continued to rise with more trains passing, there was no divergence between private and social costs.

SP/DSE/I/18

In which of the following situations will there be NO divergence between private and social costs / benefits?

- A. A chemical factory upstream pollutes a river without considering the damage inflicted upon fish farmers downstream.
- B. A diner smokes in a restaurant without considering the effects on people nearby.
- C. A farmer raises some sheep which damage the crops of nearby farmland owned by the farmer.
- D. A pianist plays the piano at home and receives no payment from his neighbour who enjoys the music.

2013/DSE/I/18

Suppose the market equilibrium output level is greater than the efficient output level. It implies that

- A. total social surplus will increase if output is reduced towards the efficient output level.
- B. at the equilibrium output level, marginal benefit is greater than marginal cost.
- C. the sum of consumer surplus and producer surplus is maximized as the market output level satisfies the equilibrium condition.
- D. government intervention is necessary to reduce output as there is over-production.

2013/DSE/I/20

A beekeeper keeps the bees for their honey. The bee-keeping benefits the farmers in the orchard nearby because the bees help pollinate the fruits in the orchard. Which of the following statements about this case is correct?

- A. There will be a divergence between private and social benefits and the honey is over-produced.
- B. The beekeeper should be subsidized by the government to improve economic efficiency.
- C. If the value of honey is higher than the value of fruits, there is no need for the government to intervene to correct the externality.
- D. There will be no divergence between private and social benefits if the orchard nearby is owned by the beekeeper.

2014/DSE/I/18

Korean pop singers are becoming more popular in Hong Kong. To these singers, which of the following are their private benefits?

- (1) the rise in royalties from the sales of their albums
 - (2) the rise in sales revenue of Korean-designed fashion
 - (3) the rise in the number of tourists visiting Korea
 - (4) the rise in revenue shared from their concerts held in Hong Kong
- A. (1) and (3) only
 - B. (1) and (4) only
 - C. (2) and (3) only
 - D. (2) and (4) only

2014/DSE/I/19

Which of the following are ways to lessen the problem of divergence between private and social costs in Hong Kong?

- (1) to collect a sewage charge
- (2) to collect a university tuition fee
- (3) to collect a charge on electricity
- (4) to collect a levy on plastic bags
- A. (1) and (3) only
- B. (1) and (4) only
- C. (2) and (3) only
- D. (2) and (4) only

2015/DSE/I/17

Which of the following is an example of the divergence between private and social benefits?

- A. A security guard watches live concerts when he is working in a concert hall.
- B. A tourist enjoys the music performed by a singer on the street and he does not pay for that.
- C. A resident living next to a karaoke lounge cannot fall asleep because of the noise.
- D. After the removal of the airport, the market value of the residential flats in that region increases.

2018/DSE/I/23

Recently, unmanned aircraft systems have become very popular but using those aircrafts in urban areas has caused noise and safety concerns. The Civil Aviation Department is considering regulating the manufacture, import and sale of unmanned aircraft systems.



It is an example of divergence between private and social _____ and the _____ solution is being considered to reduce the divergence.

- A. costs market
- B. costs government
- C. benefits market
- D. benefits government

2020/DSE/I/21



The White Moose Café
May 5 at 1:31 AM · 📍

...

****SCREAMING CHILD TAX****

For the benefit of both staff and customers alike, we have taken the decision to impose a tax on children who scream. If we notice that your child is screaming, an automatic 15% surcharge will be applied to your bill. For every further screaming child you have in your group, an additional 10% will be added.

We trust you understand our position.

👍👍👍 6.2K

1.4K Comments 533 Shares

With reference to the above information, which of the following statements is correct?

- A. Children screaming in the café results in negative externality and all cafés should introduce similar policy to restore efficiency.
- B. The extra charge on screaming children increases the marginal private cost of the groups with screaming children.
- C. The 'screaming child tax' is a government solution to the divergence between private and social costs.
- D. The 'screaming child tax' is a kind of discrimination and should not be imposed.

2022/DSE/II/22

Which of the following will most likely be the positive external effect of the opening of the full Tuen Ma Line of MTR?

- A. The travel time is shortened for MTR passengers.
- B. The business of restaurants near the stations of MTR Tuen Ma Line rises.
- C. The problem of noise pollution worsens.
- D. The income of minibus drivers drops.

Short & Structured Questions

1985/AL/I/7

'Education benefits society, therefore if the government subsidizes education, the society will benefit more.'

- (a) Critically evaluate this statement. (It is essential to remember that the production of goods also benefits society.)
- (b) It is sometimes argued that if more people are educated, then in daily social contacts the uneducated may learn from the educated free of charge. For the sake of argument, defend government subsidization under this condition.

1986/AL/I/10

The following data show that a factory, in the process of production, pollutes and inflicts damage on the neighbouring properties.

Level of Output	Total Revenue \$	Total Private Cost \$	Total Damage to Property Owners \$
100	180	130	34
200	360	270	66
300	540	420	96
400	720	580	124
500	900	750	150
600	1 080	930	174
700	1 260	1 130	196
800	1 440	1 440	216

- (a) If the factory owner seeks to maximize his own gain and ignores the damage inflicted on his neighbours, what will be the level of output? Explain.
- (b) Assume all transaction costs are zero. If the factory owner has the right to inflict damage on his neighbours, what will be the level of output? Explain.
- (c) Again assume all transaction costs are zero. If the property owners in the neighbourhood have the right to restrain the factory from polluting, what will be the level of output? Explain.

1988/AL/I/7(a)

The Hong Kong Cross-Harbour Tunnel is often congested and takes time to cross. Time is valuable. In principle, it is possible to raise the toll to reduce or even eliminate the traffic congestion.

If congestion occurs, explain why a tunnel user will generate a divergence between private and social costs, and why an increase in toll charges will reduce or eliminate this divergence.

2005/AL/I/7(a)

The Chinese government subsidizes the training of promising athletes. Some of the athletes have become sports stars, such as winners of Olympic gold medals.

On what economic ground could this subsidization in training athletes be justified?

2007/AL/I/7(a)

The Hong Kong government has imposed a strict and severe no-smoking law, prohibiting smoking in most public areas. Milton Friedman said that each and every individual should have the freedom to harm himself, including committing suicide; however, in doing so he may inflict damage on others and create a social cost problem. One who smokes cigarettes enjoys smoking, but may inflict damage on the health of others nearby.

How would you assess the social gains and costs of our new no-smoking law? Can anyone be certain that the social gains are greater than the social costs?

2008/AL/I/7(b)

Large shopping centers everywhere tend to charge much lower rents for anchor stores, or stores considered as big draws attracting visitors or shoppers to the centers, than ordinary retailers.

Fixed rents per square foot charged to anchor stores are often only small fractions of the fixed rents charged to smaller shops not noted for their drawing powers.

The anchor stores are drawing shoppers, which benefits the non-anchor stores. In what sense is there a divergence between private and social costs? In what sense does this divergence not exist?

2015/DSE/II/12(d)

Research shows that the "Individual Visit Scheme" of mainland tourists has brought economic benefits to Hong Kong. However, more and more Hong Kong residents have called on the government to reduce the number of mainland tourists to Hong Kong under the Scheme.

Some residents complained that the large number of tourists overloaded the public transportation system, resulting in a large increase in travel time for all passengers. Explain, in terms of externality, why inefficiency may exist when the transportation system is overcrowded. (4 marks)

2016/DSE/II/4

These years many customers like to take photos of the food they have in restaurants and share them on social media. However, some high-end restaurants do not allow their customers to take photos. Explain how the "photo-taking-and-sharing" behaviour of the customers may involve externality. (3 marks)

2017/DSE/II/4

Pokemon Go is a popular game for which gamers need to move around different places with their smartphones to catch the cyber-pets. Its developer has placed hotspots in different areas where many gamers are attracted to gather. Some restaurants in remote areas found their business sharply increased as they are located near these hotspots.

(a) Based on the above information, explain, in terms of externality, how the placing of the hotspots may result in inefficiency. (3 marks)

(b) How may such a problem of inefficiency be relieved by market? (2 marks)

2019/DSE/II/4

Getting vaccinated is an effective way to prevent infection and protect oneself from flu viruses. Using the concept of externality, explain why the government provision of subsidy on Seasonal Influenza Vaccination (接種季節性流感疫苗) may improve economic efficiency. (4 marks)

Short & Structured Questions

2022/DSE/II/10(a-b)

The Olympic Games attract billions of international television viewers to watch the competitions among thousands of top athletes.

- (a) Some restaurants in shopping malls find their businesses increase sharply when the malls broadcast Olympic events. Explain how the broadcast of Olympic events may result in a divergence between private and social benefits. (3 marks)
- (b) Suppose the divergence in part (a) does exist. How can market mechanism help reduce this divergence? (2 marks)

2024/DSE/II/2

Many industrial activities emit carbon dioxide, intensifying global warming and potentially bringing hazards to all human beings,

- (a) In terms of externality, explain why the emission of carbon dioxide by industrial activities may result in economic inefficiency. (4 marks)
A system of carbon credits is introduced by the governments in some countries. Carbon credits are permits for emitting carbon dioxide. Producers must purchase carbon credits from the government in order to engage in production activities that emit carbon dioxide.
- (b) Explain how the introduction of the system of carbon credits may improve economic efficiency. (2 marks)

2025/DSE/II/3

A tutorial teacher shares his short teaching videos on social media to the public free of charge. Many people benefit from his videos without paying him anything.

Explain why externality is **NOT** necessarily involved in the above situation. (3 marks)